

SQL MINI PROJECT

TASKSHEET

Data Modeling and Basic Queries

- **Task 1:** Create an ER diagram for the Amazon Fresh database to understand the relationships between tables (e.g., Customers, Products, Orders).
- **Task 2:** Identify the primary keys and foreign keys for each table and describe their relationships.
- **Task 3:** Write a query to:
Retrieve all customers from a specific city.
Fetch all products under the "Fruits" category.

Data Definition Language (DDL) and Constraints

- **Task 4:** Write DDL statements to recreate the Customers table with the following constraints:
 - CustomerID as the primary key.
 - Ensure Age cannot be null and must be greater than 18.
 - Add a unique constraint for Name.

Data Manipulation Language (DML)

- **Task 5:** Insert 3 new rows into the Products table using INSERT statements.
- **Task 6:** Update the stock quantity of a product where ProductID matches a specific ID.
- **Task 7:** Delete a supplier from the Suppliers table where their city matches a specific value.

SQL Constraints and Operators

- **Task 8: Use SQL constraints to:**
 - Add a CHECK constraint to ensure that ratings in the Reviews table are between 1 and 5.
 - Add a DEFAULT constraint for the PrimeMember column in the Customers table (default value: "No").

Clauses and Aggregations

- **Task 9: Write queries using:**
- WHERE clause to find orders placed after 2024-01-01.
- HAVING clause to list products with average ratings greater than 4.
- GROUP BY and ORDER BY clauses to rank products by total sales.

Task 10: Identifying High-Value Customers

Scenario:

Amazon Fresh wants to identify top customers based on their total spending. We will:

1. Calculate each customer's total spending.
2. Rank customers based on their spending.
3. Identify customers who have spent more than ₹5,000.

Dashboard Planning and Data Preparation

Task 11: Design an Executive Dashboard

- Create business performance metrics to provide a summarized view of Amazon Fresh's overall sales and operational performance.

Display KPI Cards for:

- Total Revenue
- Total Order Amount
- Total Orders
- Total Customers
- Total Quantity Sold
- Total Products
- Total Discount Amount
- Total Reviews
- Average Rating

Task 12: Analyze Category Sales Performance

- Compare stock availability across different products.
- Evaluate sales performance against the business target.
- Compare customer ratings across different products.

- Compare discount distribution across different products.

Task 13: Analyze Product Inventory & Sales Performance

- Compare quantity sold across different product categories.
- Compare revenue generated by different product categories.
- Compare discount amounts across different product categories.
- Evaluate the effectiveness of category-wise discount strategies.

Task 14: Analyze Customer Demographics & Purchasing Behaviour

- Compare customer purchases across different age groups.
- Analyze customer purchasing behaviour based on gender.
- Evaluate customer demographics to understand buying patterns.
- Support customer segmentation and marketing strategies.

Task 15: Analyze Regional Sales & Customer Satisfaction

- Compare revenue generated across different states.
- Compare average customer ratings across product categories.
- Evaluate regional sales performance and customer satisfaction.
- Support regional growth and customer experience improvement strategies.

Interactive Dashboard Features

Task 16: Implement Interactive Dashboard Features

Add slicers for:

- Product Category
- Age Group
- Gender
- Implement page navigation using buttons.

Dashboard Formatting

Task 17: Dashboard Design & Theme Implementation

- Designed a professional Amazon Fresh dashboard with a consistent theme across all pages.
- Applied custom dashboard backgrounds, icons, and color schemes.
- Formatted KPI cards, charts, slicers, and buttons for a clean layout.
- Maintained consistent fonts, spacing, and alignment throughout the report.

Business Insights

Task 18: Business Performance & Insight Analysis

- Highest Revenue Category
- Best Performing State
- Total Revenue Performance
- Lowest Revenue Category

Navigation and User Experience

Task 19: Interactive Navigation & User Experience

- Created a Home page for report navigation.
- Implemented page-to-page navigation using navigation buttons.
- Used bookmarks to switch between different chart views within the same report page.
- Added slicers for Product Category, State, Gender, and other filters.
- Enabled cross-filtering between visuals for interactive analysis.

Executive Summary

Task 20: Executive Business Dashboard Development

- Developed an executive dashboard summarizing key business KPIs.
- Displayed Total Revenue, Total Orders, Total Customers, and Total Quantity Sold.
- Presented monthly sales trends, category performance, and sales target achievement.
- Summarized customer, product, and regional performance using interactive visualizations.

Final Dashboard Presentation

Task 21: Business Recommendation & Decision Support Dashboard

- Developed a recommendation dashboard based on analytical findings.
- Used KPI cards, a decomposition tree, and opportunity analysis to evaluate business performance.
- Recommended inventory improvements for high-demand products.
- Suggested discount optimization and promotional strategies.
- Proposed actions to improve customer satisfaction, product performance, and regional sales growth.